



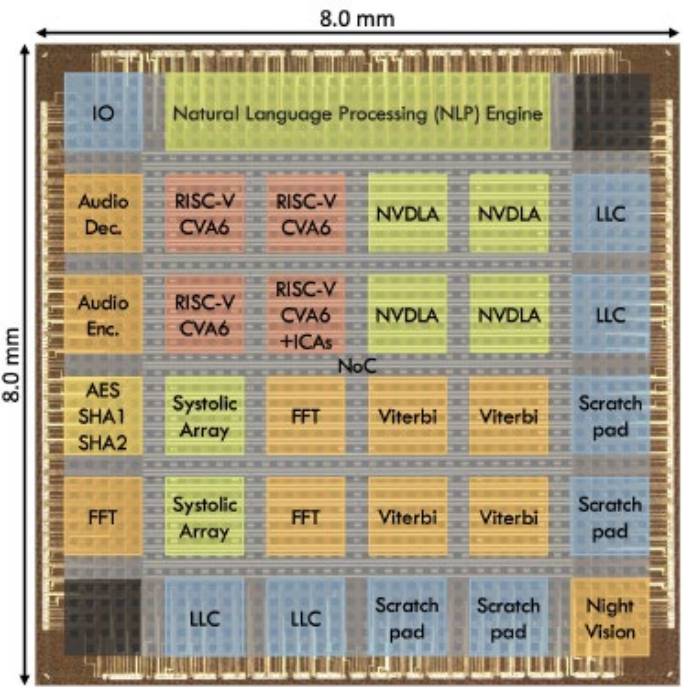
From ESP to OSCAR: Building Open Platforms for Research, Education, and Collaboration

Luca P. Carloni



Open-Source EDA Birds-of-a-Feather: Open-Source
EDA, Data and Collaboration Summit at DAC 2026
Long Beach - July 28, 2026

An Exciting Open-Source Project: the EPOCHS-1 Chip



Technology	12nm FinFET
Area	64mm ²
#IOs	340
Power Domains	23
Clock Domains	35
Power	83mW – 4.33W
Total SRAM	8.4MB
Max Frequency Range	680MHz – 1.6GHz
Example Application Domain	Collaborative Autonomous Vehicles

14.5 A 12nm Linux-SMP-Capable RISC-V SoC with 14 Accelerator Types, Distributed Hardware Power Management and Flexible NoC-Based Data Orchestration

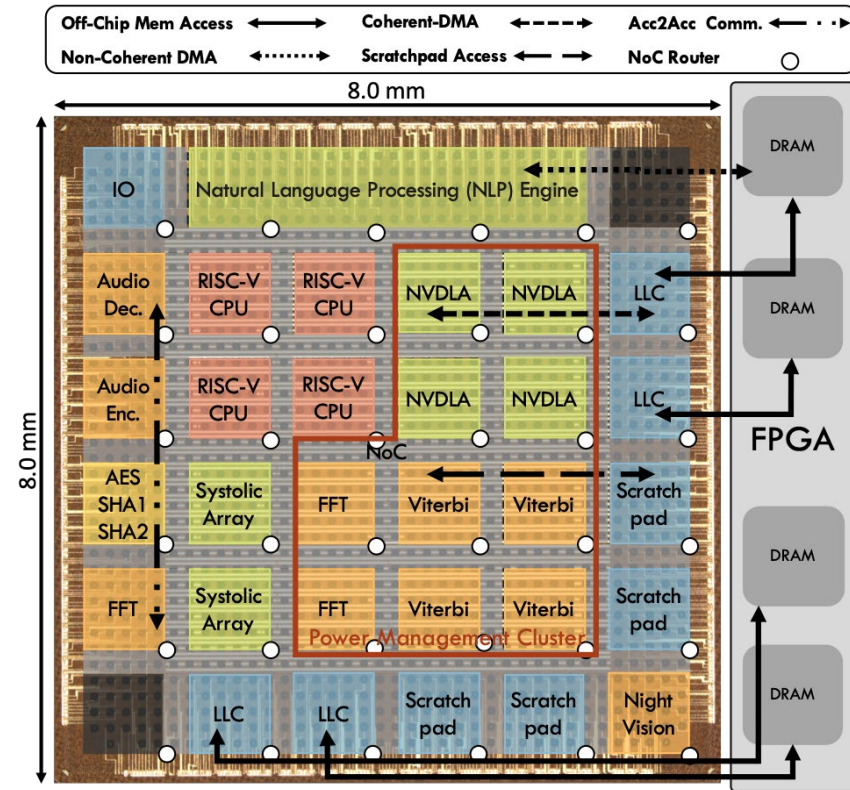
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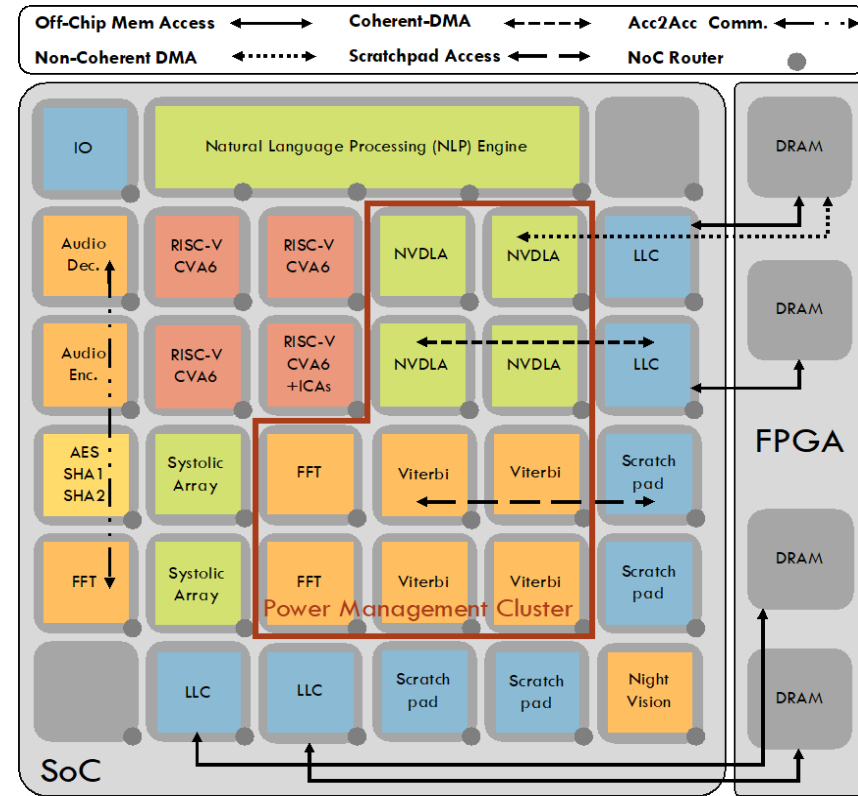
The EPOCHS-1 SoC: Chip Highlights

- 64 mm² SoC designed in 12 nm FinFET
- 35 clock domains; 23 power domains
- 8.4 MB on-chip SRAM memory
- Tile-based SoC architecture



The EPOCHS-1 SoC: Chip Highlights

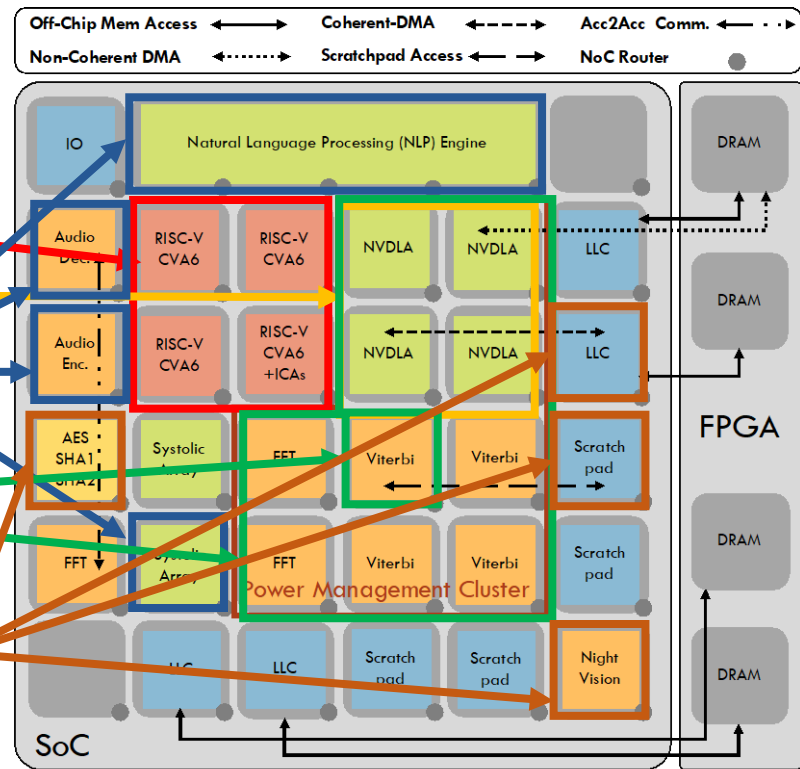
- 64 mm² SoC designed in 12 nm FinFET
- 35 clock domains; 23 power domains
- 8.4 MB on-chip SRAM memory
- Tile-based SoC architecture
- 34 tiles connected by a 6-plane, 2-D mesh NoC
- The 74 Tbps NoC provides flexible orchestration of data
- 23 accelerators of 14 different types
- 10 accelerators compose a cluster demonstrating a novel distributed hardware power management (DHPM) scheme
- Designed by a small team of PhD students, postdocs, and industry researchers in 3 months with ESP, our open-source platform for agile SoC design



The EPOCHS-1 SoC: Sources of OSH IPs

• Sources of Open-Source Hardware IPs:

- 4 RISC-V CVA6 cores from ETH Zurich/OpenHW Group
- 4 NVIDIA Deep Learning Accelerators
- 4 Accelerators designed at Harvard
- 1 Accelerator and Power Management designed at IBM Research
- 3 Accelerators, Memory Hierarchy, and Network-on-Chip designed at Columbia



ESP : An Open-Source Platform for SoC Design

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ESP

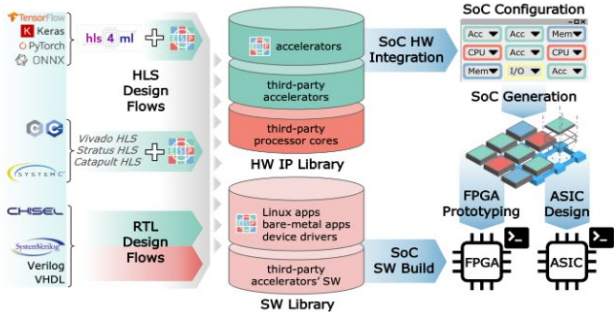
the open-source SoC platform

esp.cs.columbia.edu



The ESP Vision

ESP is an open-source research platform for heterogeneous system-on-chip design that combines a scalable tile-based architecture and a flexible system-level design methodology.

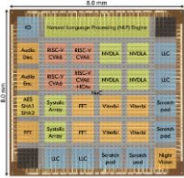


ESP provides three accelerator flows: RTL, high-level synthesis (HLS), machine learning frameworks. All three design flows converge to the ESP automated SoC integration flow that generates the necessary hardware and software interfaces to rapidly enable full-system prototyping on FPGA.

Overview



Latest Posts



ESP at ISSCC!

Check out our second chip based on ESP, the open-source SoC platform.

[Read more](#)

Published: Mar 16, 2024



Release 2024.1.0

A new GitHub Release

The Concept of Platform

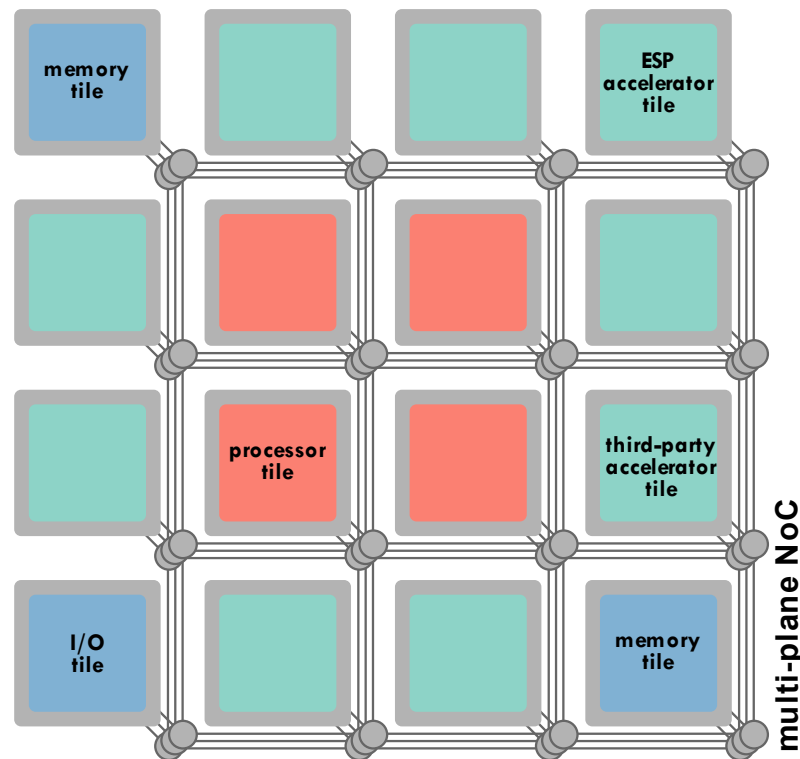
- Innovation in SoC architectures and their design methodologies is needed to promote design reuse and collaboration
 - Architectures and methodologies must be developed together
- ***Platform = architecture + methodology***
 - An SoC architecture enables design reuse when it simplifies the integration of many components that are independently developed
 - An SoC methodology enables design collaboration when it allows designers to choose the preferred specification languages and design flows for the various components
- An effective combination of architecture and methodology is a platform that maximizes the potential of open-source hardware
 - by scaling up the number and type of components that can be integrated in an SoC and by enhancing the productivity of the designers who develop and use them



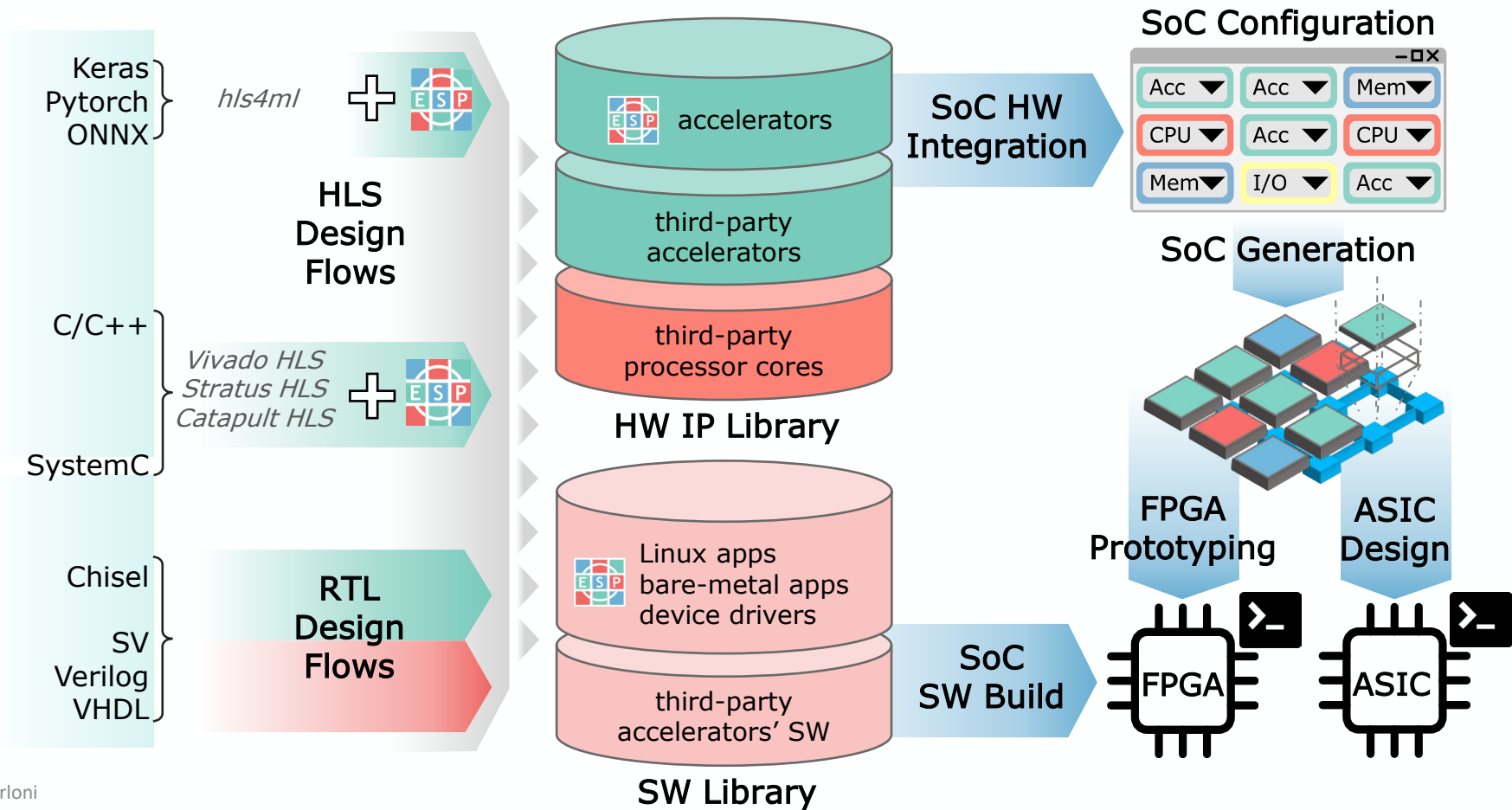
ESP Architecture

- RISC-V Processors
- Many-Accelerator
- Distributed Memory
- Multi-Plane NoC

The ESP architecture implements a **distributed** system, which is **scalable**, **modular** and **heterogeneous**, giving processors and accelerators similar weight in the SoC

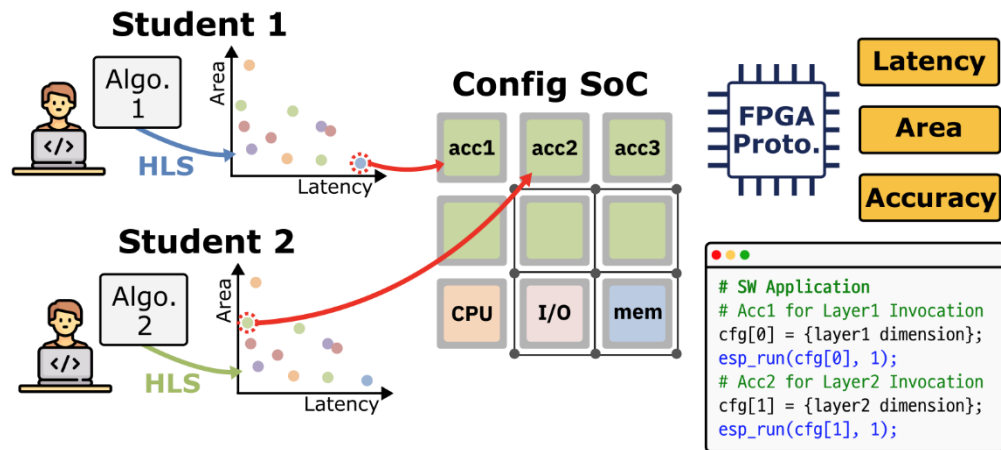


The ESP Vision: Domain Experts Can Design SoCs



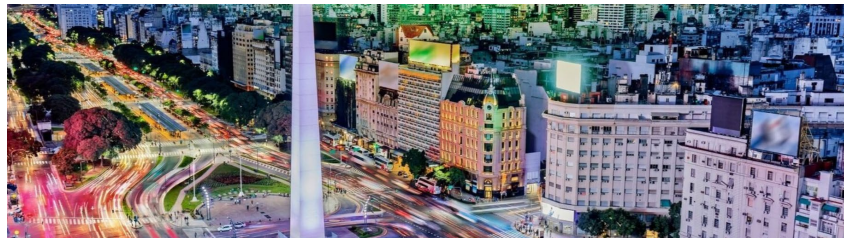
Teaching Collaborative SoC Design with ESP

- In the CSEE-4868 final project, students:
 - design multiple alternative Pareto-optimal instances of a hardware accelerator with high-level synthesis
 - build an SoC by combining their accelerator design with their selections of other accelerators designed by their fellow students
 - implement an FPGA-based prototype of the SoC, and
 - evaluate their design by running an AI software application that invokes the accelerators



The OSCAR Workshop Series

<https://oscar-workshop.github.io/>



• 2024 - Buenos Aires, Argentina



• 2023 - Orlando, FL

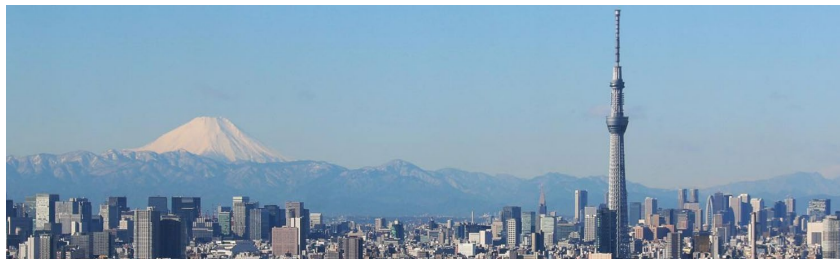


• 2022 - New York, NY

- *The goal of OSCAR is to bring together a community of researchers from academia, industry and government labs who are interested in in developing and sharing **open-source hardware and software** for the design of next-generation computer systems*



• 2026 Raleigh, NC



• 2025 Tokyo, Japan



In Summary

- **Open-Source Hardware platforms play a unique role for the development of Open-Source EDA**
 - Promote collaboration across universities, industry and government labs
 - Enable the creation of complex prototypes via design reuse that can drive tool development and innovation
 - Simplify reproducibility and benchmarking
 - Support workforce development by allowing the creation of exciting project that inspire talented students
- **The Age of AI offers unique opportunities to build virtuous cycles between Open-Source Hardware and Open-Source EDA**
 - Let's work together!

